

Part A – Theoretical Foundation (Short Notes & Explanation)

1. What is inferential statistics?

Inferential statistics is like being a data detective. It's the process of taking a small group (a sample) and using it to make smart guesses or conclusions about a much larger group (the population). Since we can't always survey everyone in the world, we use these tools to see if our findings are likely true for everyone.

2. What is hypothesis testing and its components?

Hypothesis testing is a formal statistical method used to make decisions about population parameters based on sample data. It involves comparing two competing claims to determine which one is better supported by the evidence. It is a process that helps you determine whether an assumption (hypothesis) about a population is likely to be true or if the results happened by pure chance.

3. Explain confidence interval and critical value.

A Confidence Interval is a range of values, derived from sample data, that is likely to contain the true population parameter at a specified confidence level. It provides an estimate of the uncertainty or margin of error associated with a sample statistic. Instead of giving a single number as a guess, it provides a "likely range" (like 95% or 99%) where the real answer is expected to be found.

4. Define p-value.

The p-value is the probability of obtaining test results at least as extreme as the results actually observed, under the assumption that the null hypothesis is correct. It serves as a measure of the strength of the evidence against the null hypothesis.

5. Differentiate Type I and Type II errors.

Type I and Type II errors are the two types of mistakes that can occur during hypothesis testing when making a decision about the null hypothesis. A **Type I error** happens when you reject a true null hypothesis, while a **Type II error** happens when you fail to reject a false null hypothesis.

6. Brief descriptions of z-test, t-test, chi-square test, and ANOVA test.

These are specific statistical tools used to compare datasets and find relationships between variables based on different data conditions.

- **Z-test:** A test used to compare means when the sample size is large ($n > 30$) and the population variance is known.
- **T-test:** A test used to compare the means of two groups when the sample size is small or the population variance is unknown.
- **Chi-square Test:** A test used to determine if there is a significant association or pattern between two categorical variables.
- **ANOVA (Analysis of Variance):** A test used to compare the means of three or more independent groups simultaneously to see if at least one is significantly different.

7. What is Covariance?

Covariance is a measure of the directional relationship between two random variables to see how they change together in a linear fashion. A positive covariance indicates that both variables tend to increase or decrease at the same time, while a negative covariance indicates that one variable tends to increase as the other decreases.

8. What is Correlation?

Correlation is a standardized measure that describes both the strength and the direction of the linear relationship between two variables. Unlike covariance, correlation is scaled between -1 and +1, where a value of +1 indicates a perfect positive relationship, -1 indicates a perfect negative relationship, and 0 indicates no linear relationship.